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Exchange Functional PBE Convergence Test

Table 1: DFT Embedding with PBE Exchange: Energy Analysis

Molecule	Active Space	Reference DFT Energy (Ha)	Final Ground-State Energy (Ha)
H_2	(2e, 2o)	−1.161905	−1.182605
H_2O	(2e, 6o)	−76.320335	−76.341891
H_2O	(8e, 6o)	−76.320335	−76.353242
O_2	(2e, 6o)	−150.174613	−150.136870
O_2	(8e, 6o)	−150.174613	−150.300681

Table 2: DFT Embedding with PBE Exchange: Convergence Analysis

Molecule	Active Space	Converged at Iteration	Energy Tolerance (Ha)	Wall Time(sec)
H_2	(2e, 2o)	9	$7.669e{-08}$	61
H_2O	(2e, 6o)	9	$2.321e{-07}$	1081
H_2O	(8e, 6o)	9	$2.54e{-07}$	5, 516
O_2	(2e, 6o)	4	$3.599e{-07}$	141
O_2	(8e, 6o)	27	$5.873e{-07}$	9217

Exchange Functional B3LYP Convergence Test

Table 3: B3LYP Exchange: Energy Analysis (alpha_init = 0.6)

Molecule	Active Space	Reference DFT Energy (Ha)	Final Ground-State Energy (Ha)
H_2	(2e, 2o)	−1.175477	−1.190061
H_2O	(2e, 6o)	−76.407023	−76.425277
H_2O	(8e, 6o)	−76.407023	−76.431931
O_2	(2e, 6o)	−150.312682	−150.220130
O_2	(8e, 6o)	−150.312682	−150.353251

Table 4: B3LYP Exchange: Convergence Analysis (alpha_init = 0.6)

Molecule	Active Space	Converged at Iteration	Energy Tolerance (Ha)	Wall Time(sec)
H_2	(2e, 2o)	9	$6.683e{-07}$	61
H_2O	(2e, 6o)	9	$9.587e{-07}$	1115
H_2O	(8e, 6o)	10	$7.01e{-07}$	6,052
O_2	(2e, 6o)	3	$7.153e{-07}$	123
O_2	(8e, 6o)	14	$5.23e{-07}$	4,727

wB97X and PBE0 Convergence Test

Table 5: wB97X Exchange: Energy Analysis (alpha_init = 0.4)

Molecule	Active Space	Reference DFT Energy (Ha)	Final Ground-State Energy (Ha)
H_2	(2e, 2o)	−1.167524	−1.175589
H_2O	(2e, 6o)	−76.386535	−76.400538
O_2	(2e, 6o)	−150.277948	−150.078457

Table 6: wB97X Exchange: Convergence Analysis (alpha_init = 0.4)

Molecule	Active Space	Converged at Iteration	Energy Tolerance (Ha)	Wall Time(sec)
H_2	(2e, 2o)	4	$7.412e{-07}$	40
H_2O	(2e, 6o)	8	$5.456e{-08}$	950
O_2	(2e, 6o)	2	$6.982e{-07}$	98

Table 7: PBE0 Exchange: Energy Analysis (alpha_init = 0.4)

Molecule	Active Space	Reference DFT Energy (Ha)	Final Ground-State Energy (Ha)
H_2	(2e, 2o)	−1.163879	−1.177360
H_2O	(2e, 6o)	−76.323911	−76.341380
O_2	(2e, 6o)	−150.160818	−150.053725

Table 8: PBE0 Exchange: Convergence Analysis (alpha_init = 0.4)

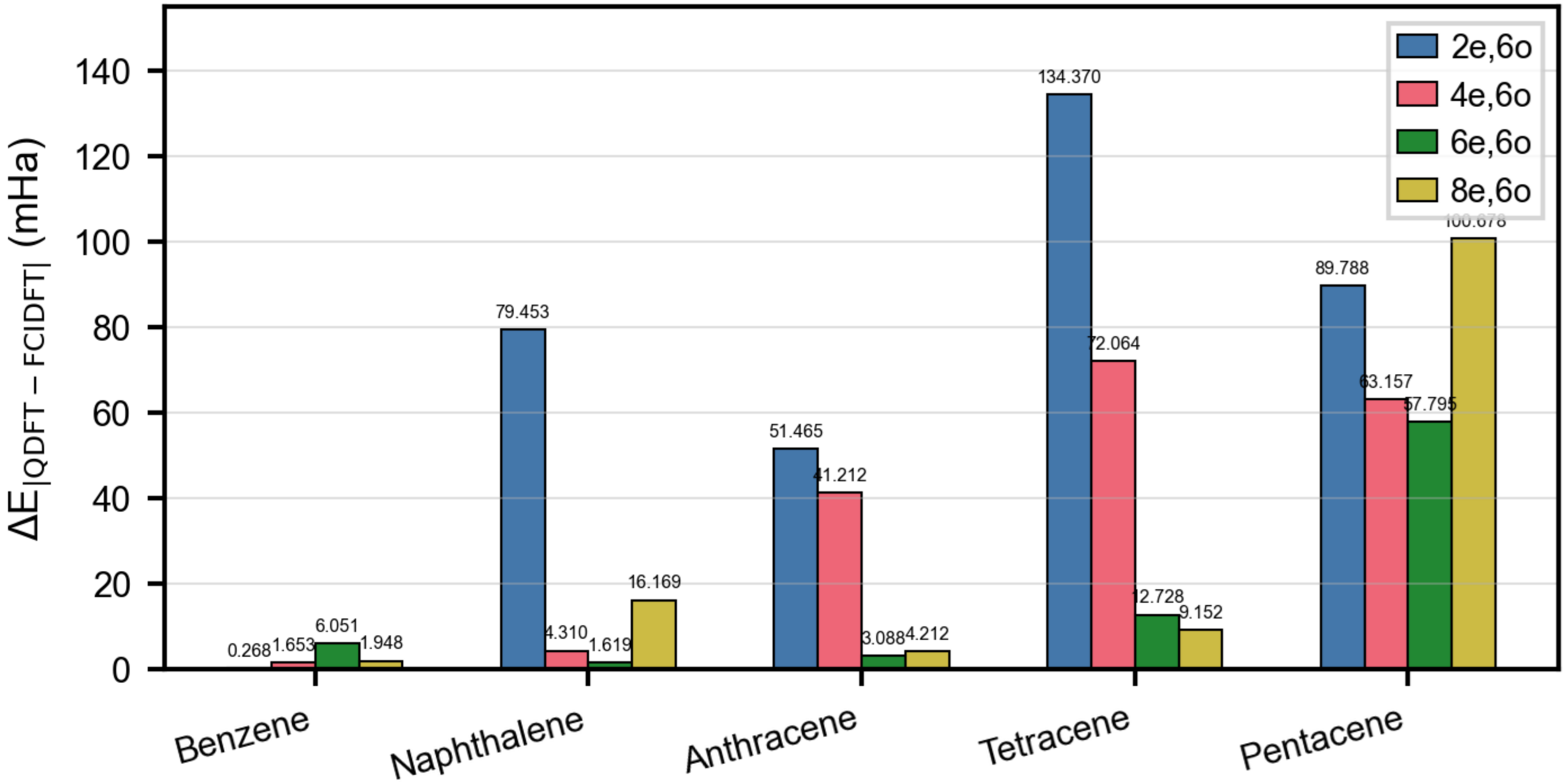
Molecule	Active Space	Converged at Iteration	Energy Tolerance (Ha)	Wall Time(sec)
H_2	(2e, 2o)	8	$7.606e{-07}$	58
H_2O	(2e, 6o)	8	$3.611e{-07}$	965
O_2	(2e, 6o)	3	$6.280e{-07}$	126

Reaction Energy Comparison Across Functionals

Functional	Active Space	Reaction Energy (Ha)
ωB97X (new)	(2e,60)	-0.371441
PBE0 (new)	(2e,60)	-0.274313
B3LYP (old)	(2e,60)	-0.250302
PBE (new)	(2e,60)	-0.181425
PBE (complex)	(2e,60)	-0.181702
LDA (new)	(2e,60)	-0.180088
B3LYP (old)	(8e,60)	-0.130489
PBE (complex)	(8e,60)	-0.040592

New -> New Simplified Damping method (Linear damping + DIIS)
Complex -> Complex damping method (Linear damping + DIIS)
Old -> Linear damping method

mHa difference B3LYP



mHa difference cam-B3LYP

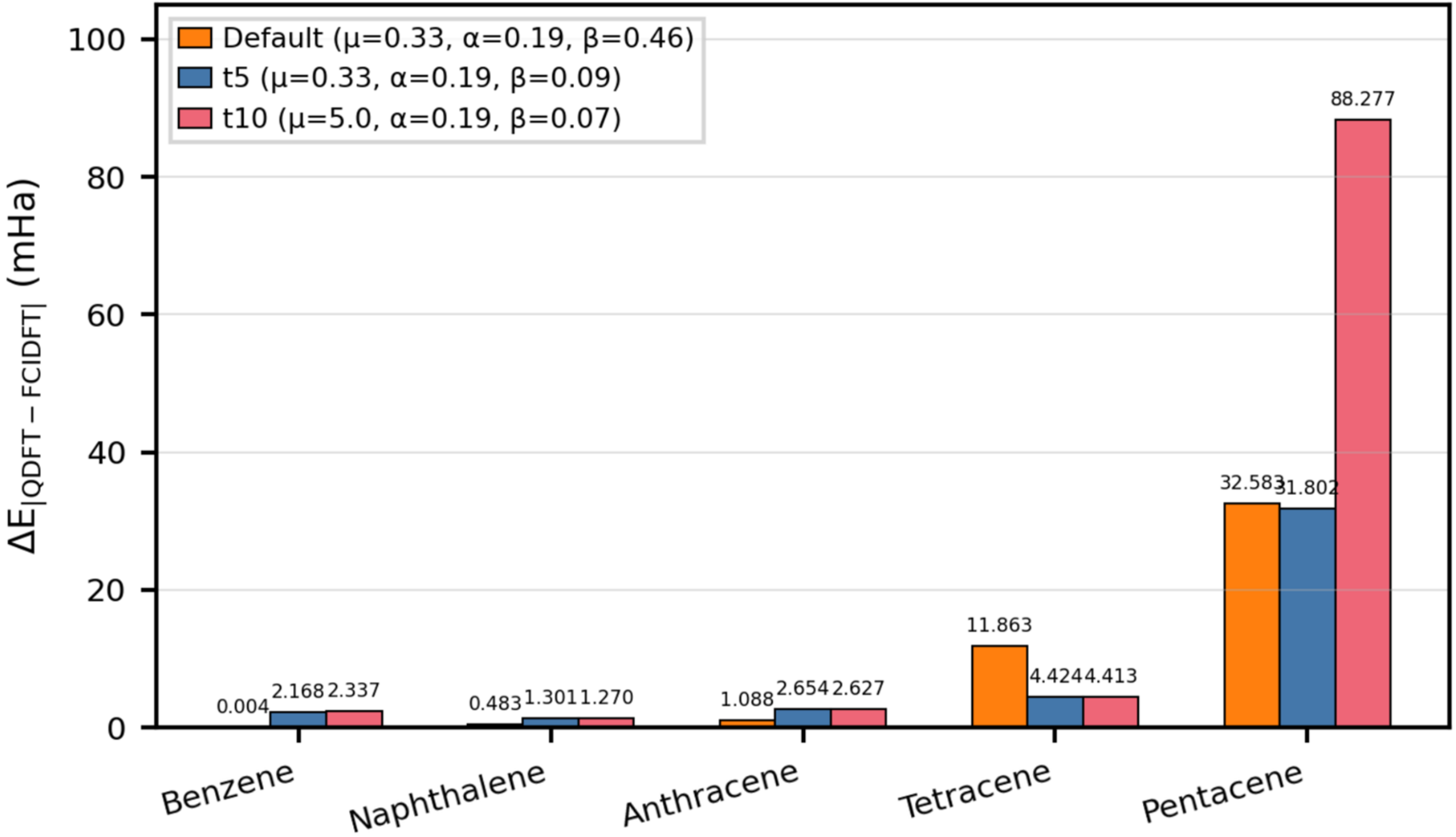
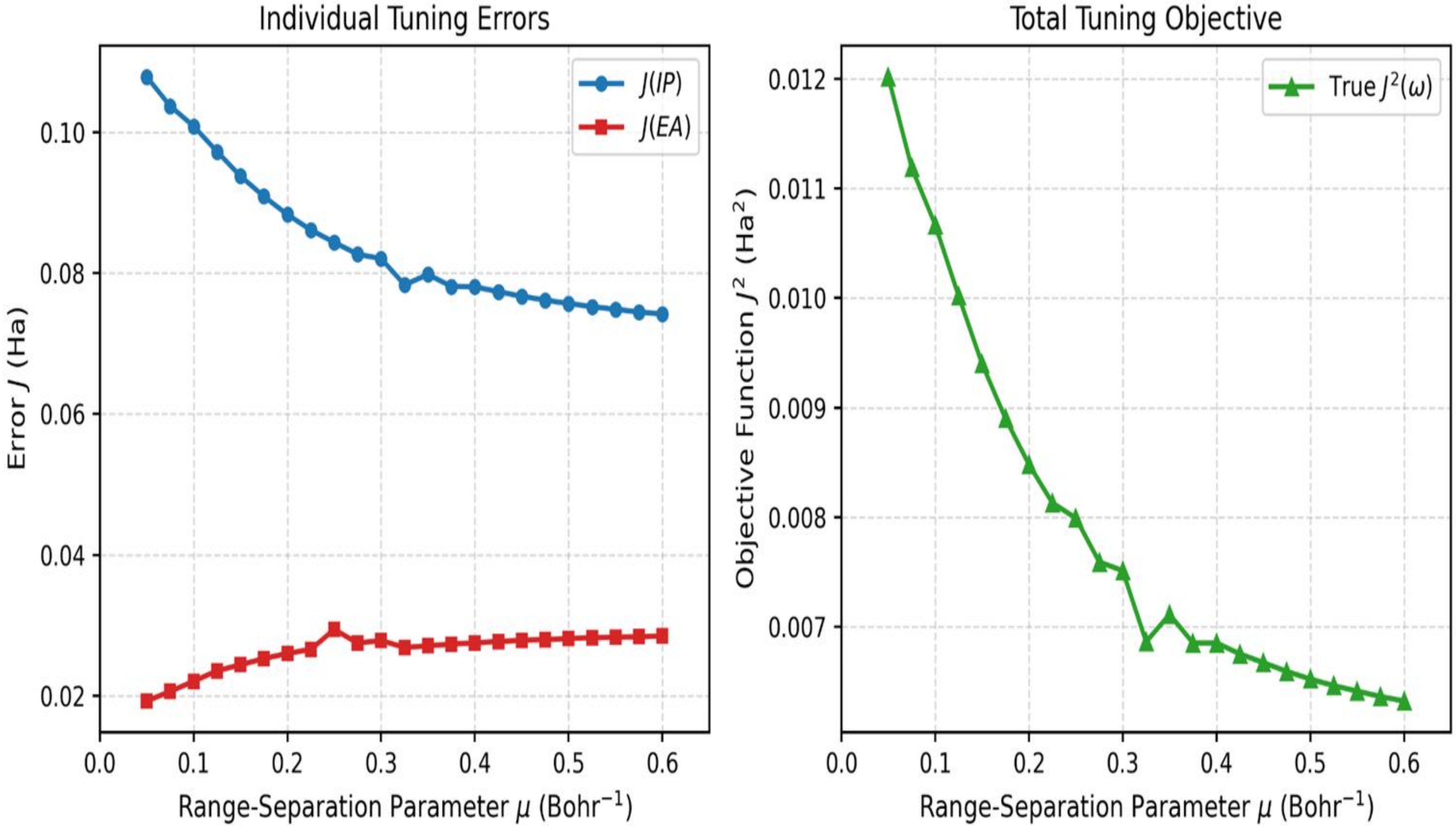


Table 7: Tuning of the range-separation parameter μ for Tetracene focusing on Electron Affinity (EA) characteristics and the final objective function J^2. Energies are in Hartrees.

μ	RKS Energy	ROKS Anion Energy	Anion HOMO	EA	$J(EA)$	True J^2
0.050	-693.34511	-693.30850	0.05584	-0.03661	0.01922	0.01202
0.075	-693.24543	-693.21012	0.05588	-0.03531	0.02057	0.01119
0.100	-693.16300	-693.12896	0.05606	-0.03404	0.02202	0.01067
0.125	-693.09447	-693.06163	0.05634	-0.03284	0.02351	0.01002
0.150	-693.03705	-693.00495	0.05648	-0.03211	0.02437	0.00940
0.175	-692.98845	-692.95707	0.05661	-0.03138	0.02524	0.00890
0.200	-692.94679	-692.91606	0.05671	-0.03073	0.02598	0.00848
0.225	-692.91061	-692.88041	0.05674	-0.03021	0.02654	0.00813
0.250	-692.87871	-692.85166	0.05644	-0.02705	0.02939	0.00799
0.275	-692.85016	-692.82086	0.05674	-0.02930	0.02744	0.00759
0.300	-692.82422	-692.79531	0.05673	-0.02891	0.02782	0.00751
0.325	-692.80033	-692.77099	0.05616	-0.02934	0.02682	0.00686
0.350	-692.77803	-692.74900	0.05610	-0.02903	0.02707	0.00711
0.375	-692.75700	-692.72823	0.05607	-0.02877	0.02730	0.00685
0.400	-692.73698	-692.70843	0.05602	-0.02855	0.02747	0.00685
0.425	-692.71778	-692.68944	0.05603	-0.02834	0.02768	0.00675
0.450	-692.69928	-692.67110	0.05602	-0.02818	0.02784	0.00667
0.475	-692.68138	-692.65334	0.05601	-0.02804	0.02796	0.00659
0.500	-692.66404	-692.63610	0.05602	-0.02794	0.02809	0.00652
0.525	-692.64723	-692.61938	0.05606	-0.02785	0.02821	0.00646
0.550	-692.63094	-692.60315	0.05610	-0.02779	0.02830	0.00641
0.575	-692.61518	-692.58742	0.05611	-0.02776	0.02834	0.00636
0.600	-692.59998	-692.57225	0.05618	-0.02773	0.02845	0.00632



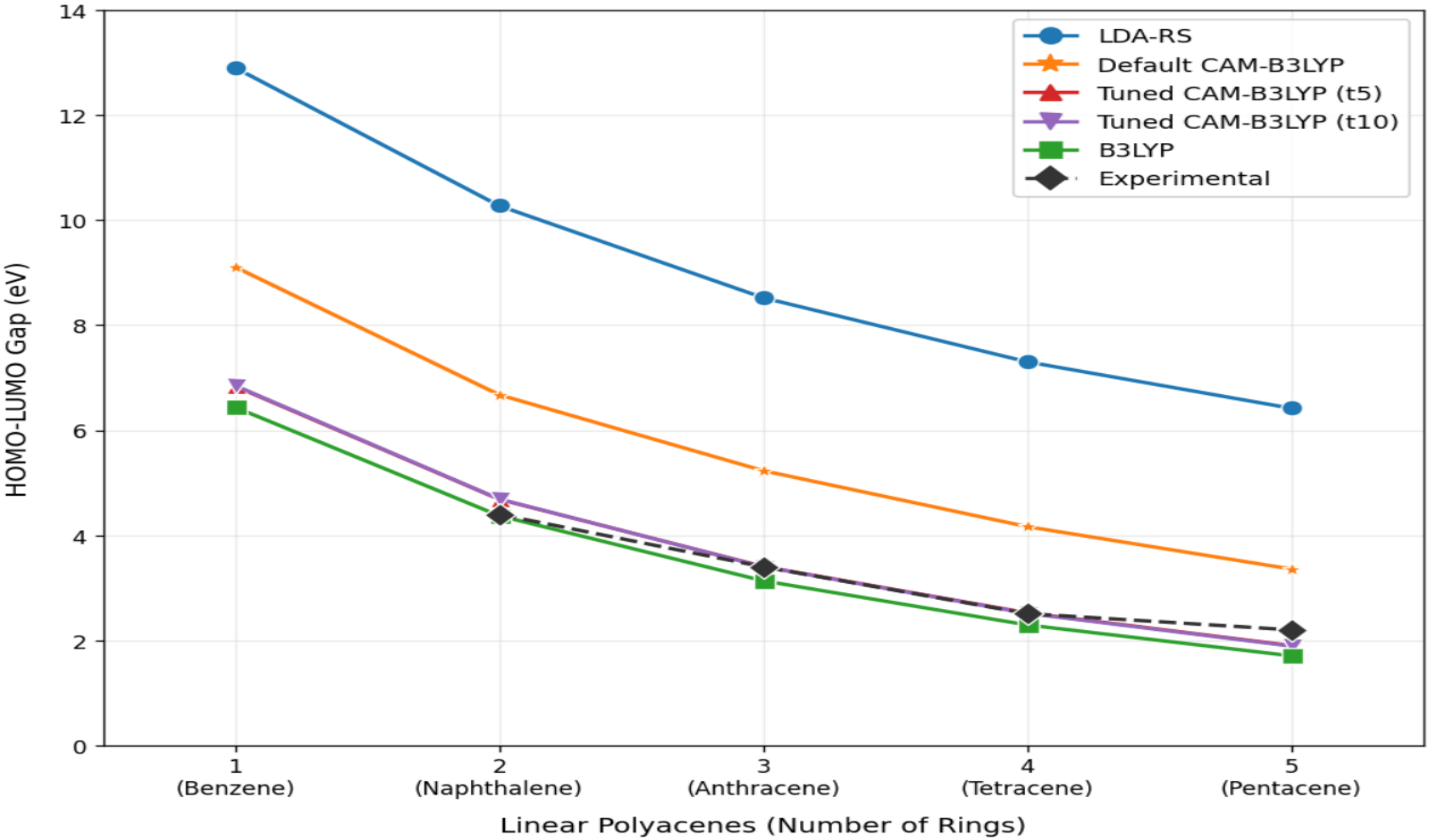
Individual Tuning Errors
Error J (Ha)
0.02
0.04
0.06
0.08
0.10
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Range-Separation Parameter μ (Bohr^{-1})
$J(IP)$
$J(EA)$
Total Tuning Objective
Objective Function J^2 (Ha^2)
0.007
0.008
0.009
0.010
0.011
0.012
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Range-Separation Parameter μ (Bohr^{-1})
True $J^2(\omega)$

Homo-Lumo Runs

Shreyas modified the code for Homo-Lumo runs and used LDA-RS exchange functional.
I continued by finding out that Homo-Lumo gap was best with B3LYP exchange functional.
But, there was a catch; mHa difference between total ground state energy with VQE+DFT and FCI run was poor for B3LYP and great for LDA-RS.

Hence used cam-b3lyp (internally range separated functional).
Modified code to tune the parameters of cam-b3lyp functional; to get best of both mHa difference and Gap.

Homo-Lumo Gap PAH Molecules



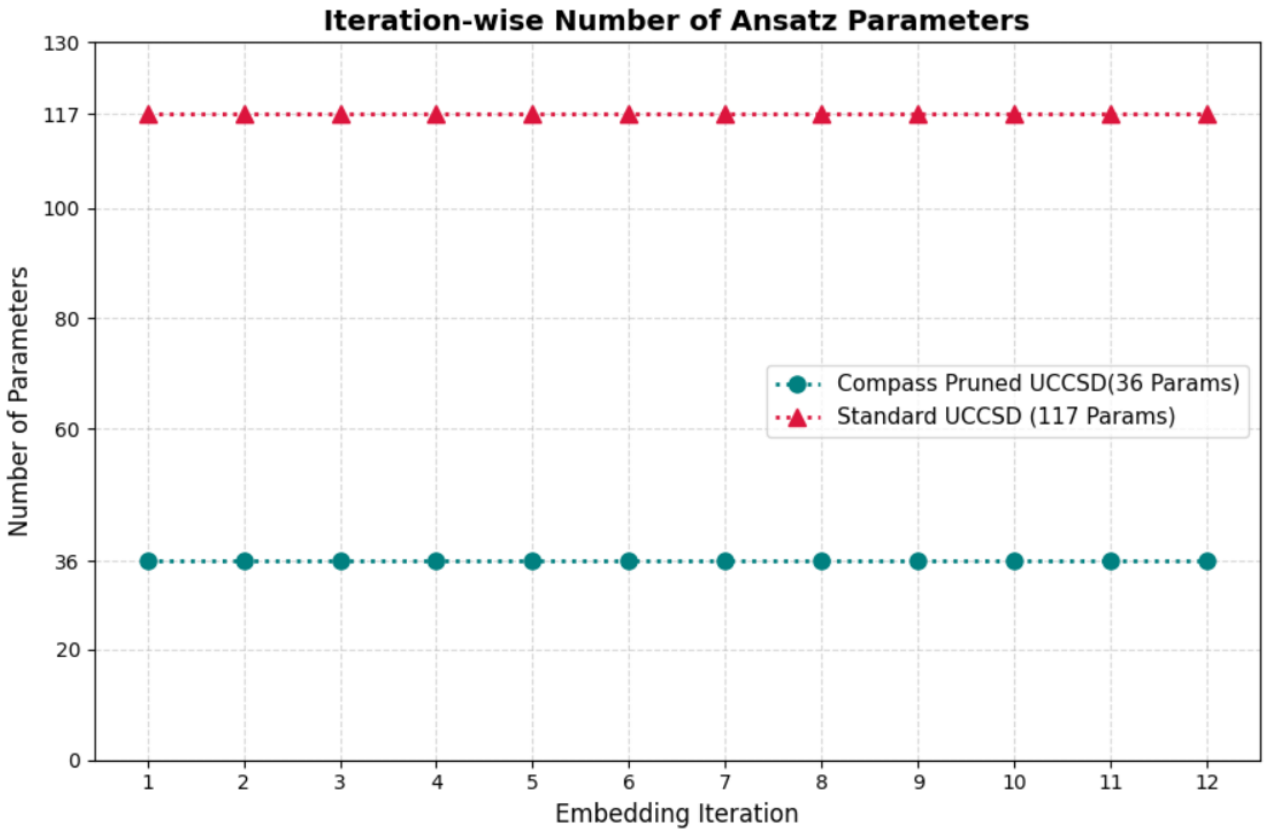
IITB Ansatz and Initial Point

Lrc-wpbe; another internally range separated functional, was found to be best for Pyridine and Porphyrin runs using UCCSD ansatz.

Modified an ansatz and initial point code patch provided by ma’am to run for Pyridine and Porphyrin (AS : 6e6o).

Found out that the code reduces the number of parameters without compromising with the accuracy, reducing the runtime.

Working on a much more modified version of the code, with includes triplets to gain more accuracy



For Pyridine 6e6o

THANK YOU